CE2007

Microprocessor-Based Systems Design

Communication Interface and Protocol - Part 2

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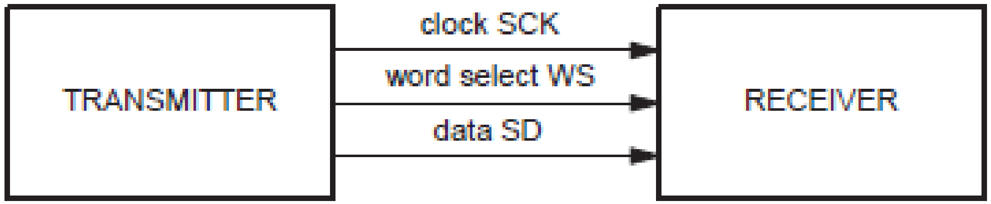
I2S

- Many digital systems have been introduced to the consumer audio market like compact disc, digital audio tape, digital sound processors, digital TV-sound, etc.
- The digital audio signals in these systems are being processed by a number of IC chips such as:
  - A/D and D/A converters
  - DSPs
  - Error Correction Chips
  - Digital Filters
  - Digital I/O Interfaces
- Standardized communication structures are vital for both the equipment and the IC manufacturer to increase system flexibility.
- I2S (Inter-IC Sound) – A serial link especially for digital audio

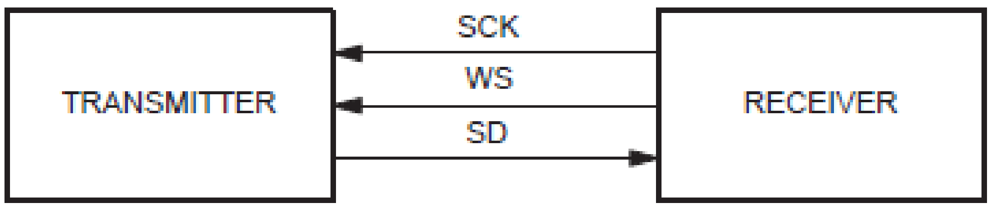
I2S – Bus Requirements

- The system only has to handle audio data.
- It consist of 3 lines
  - SCK – Clock
  - WS – Word Select
  - SD – Data
- The transmitter and receiver have the same clock signal for data transmission.
- The transmitter as the Master has to generate the bit clock, word-select signal and data.
- In a complex system, there may be several transmitters and receivers, which makes it difficult to define the Master.
- In such systems, there is usually a System Master controlling digital audio-flow between the various IC's.

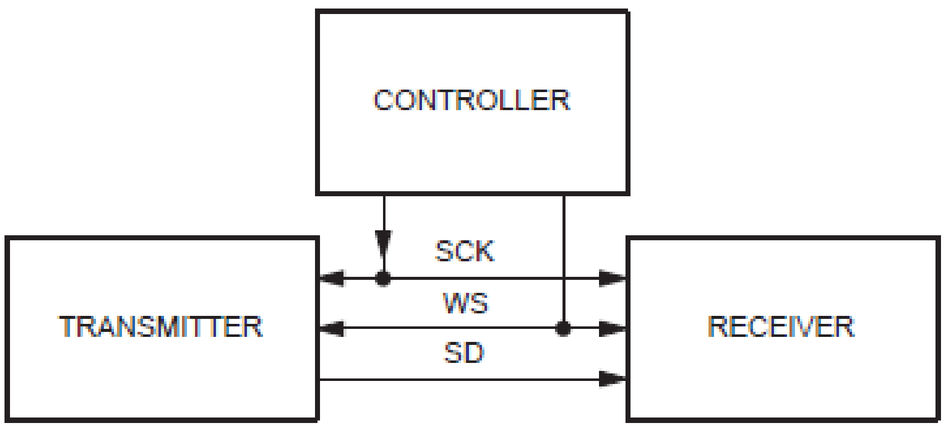
I2S – System Configurations



TRANSMITTER = MASTER



RECEIVER = MASTER



CONTROLLER = MASTER

I2S – Serial Data

- Serial data is transmitted in 2's complement with the MSB first.
- The MSB is transmitted first because the transmitter and receiver may have different word lengths.
- It isn't necessary for the transmitter to know how many bits the receiver can handle, nor does the receiver need to know how many bits are being transmitted.

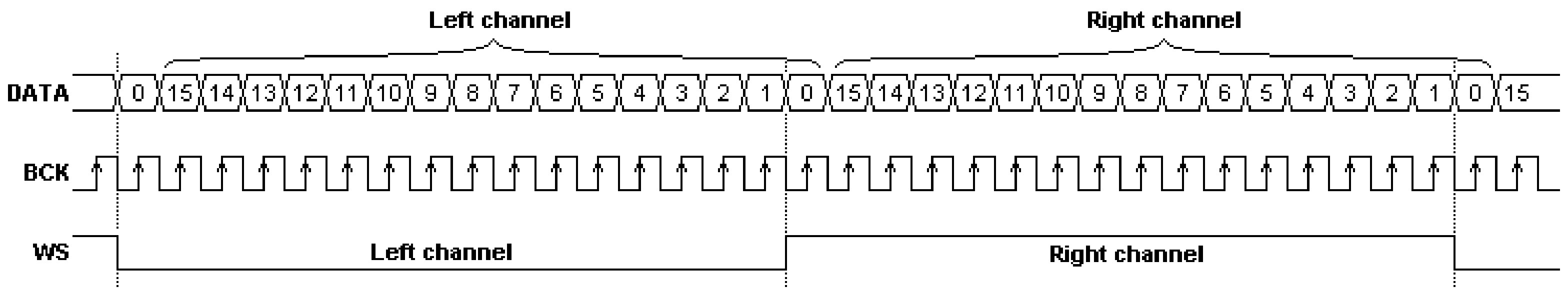
So how does the protocol handle different data packet sizes?

I2S – Serial Data

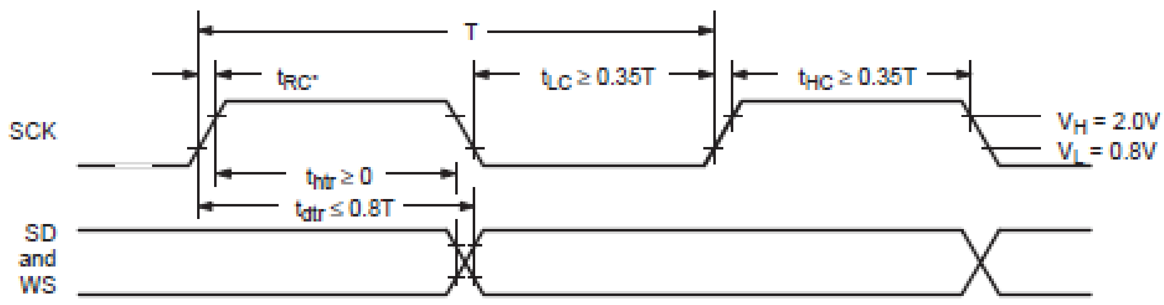
- If the receiver is sent more bits than its word length, the bits after the LSB are ignored.
- If the receiver is sent fewer bits than its word length, the missing bits are set to zero internally.
- So, the MSB has a fixed position, whereas the position of the LSB depends on the word length.

I2S – Serial Data

- The transmitter always sends the MSB of the next word one clock period after the WS changes.



- As always, timing parameters are critical and must be obeyed.



CAN – Controller Area Network

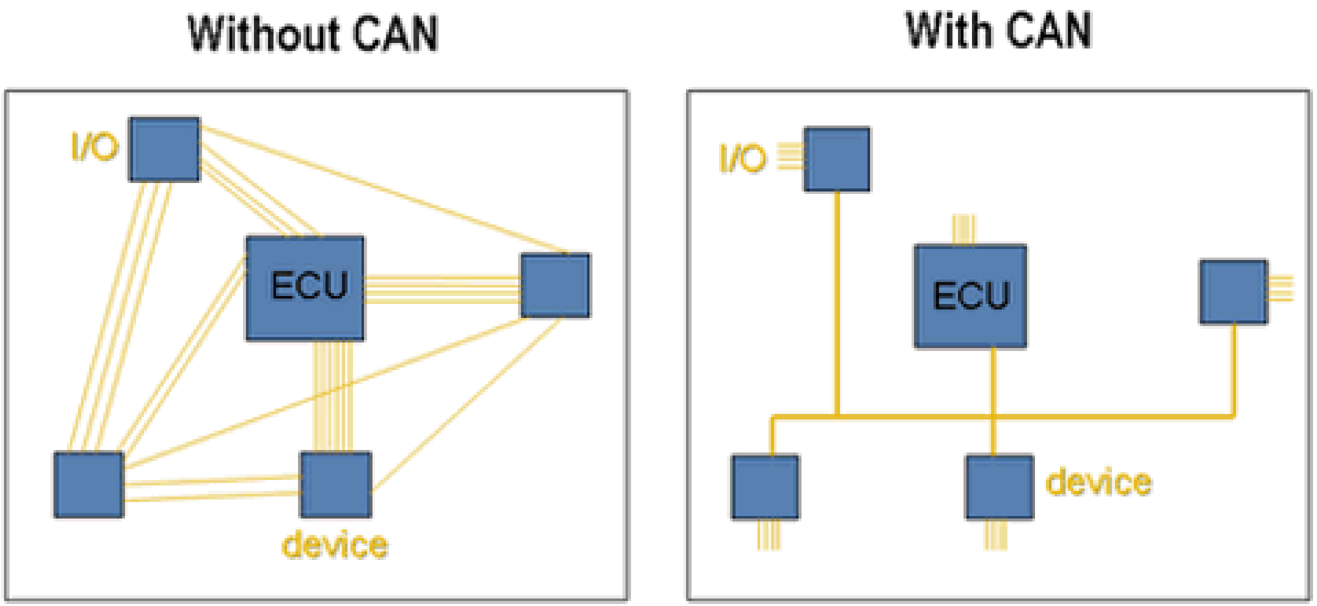
- Bosch originally developed the Controller Area Network (CAN) in 1985 for in-vehicle networks.
- Prior to that, manufacturers used point-to-point wiring systems which became impractical as the usage of in-vehicle electronics greatly increased.
- CAN, a high-integrity serial bus system for networking intelligent devices, emerged as the standard in-vehicle network.
- In 1993, the automotive industry adopted it and it became an International Standard known as ISO 11898.

CAN – Benefits

- Low-Cost, Lightweight Network
  - Provides an inexpensive, durable network that helps multiple CAN devices communicate with each other.
- Broadcast communication
  - Each device on the network has a CAN controller chip.
  - All devices on the network see all transmitted messages.
  - Each device can decide if a message is relevant or if it should be filtered.
  - Allows modifications to the network with minimal impact.
- Priority
  - Every message has a priority.
  - If two nodes try to send messages simultaneously, the one with the higher priority gets transmitted and the one with the lower priority gets postponed.
  - Non-destructive arbitration which results in non-interrupted transmission of the high priority message.
  - Network can meet deterministic timing constraints

CAN – Benefits

- Error Capabilities
  - Specification includes a Cyclic Redundancy Code (CRC) to perform error checking on each frame's contents.
  - Frames with errors are disregarded by all nodes.



CAN – Physical Layers

- CAN has different physical layers that classify certain aspects of the CAN network, such as electrical levels, signaling schemes, cable impedance, maximum baud rates, and more.
  The most common ones are:
- High-Speed CAN
  - Most common physical layer.
  - Implemented with 2 wires and allow communication at transfer rates up to 1 Mbit/s.
  - E.g. – Antilock Brake Systems, Engine Control Modules.
- Low-Speed/Fault-Tolerant CAN Hardware
  - Implemented with 2 wires and allow communication at transfer rates up to 125 Kbit/s.
  - E.g. – Opening/Closing of Doors.
- Single-Wire CAN Hardware
  - Allows transfer rates up to 33.3 Kbit/s (88.3 Kbit/2 in high-speed mode).
  - E.g. Seat and Mirror Adjusters.

CAN – Terminology

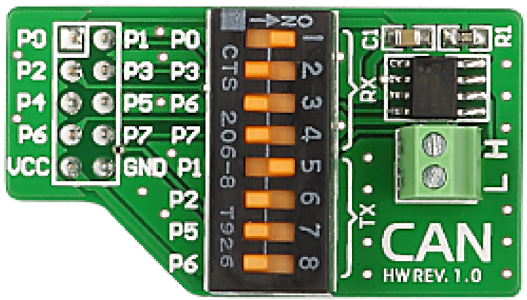
- The frame format is as follows:

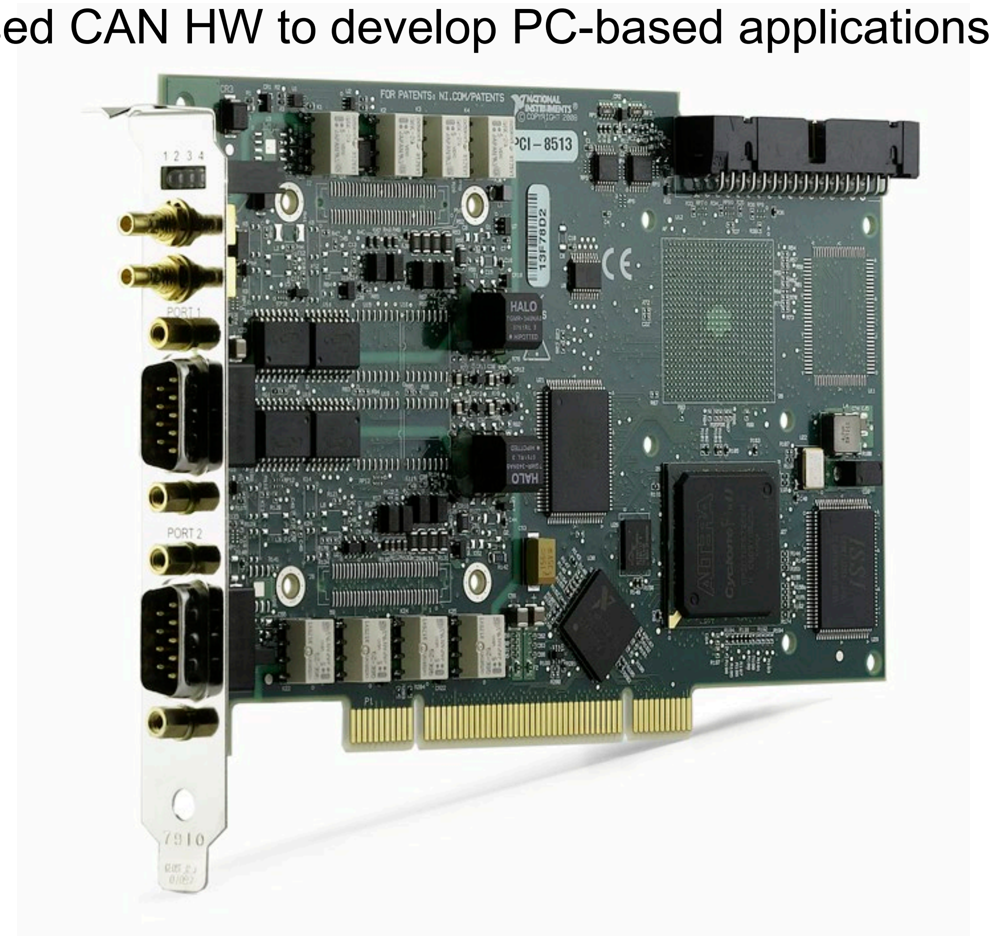
SOF	11-BIT ARBITRATION ID	SRR	IDE	18-BIT ARBITRATION ID	RTR	r0	DLC	0...8 BYTES DATA	CRC	ACK	EOF

Field Name	Length (bits)
Start of Frame (SOF)	1
Identifier A	11
Substitute Remote Request (SRR)	1
Identifier Extension bit (IDE)	1
Identifier B	18
Remote Transmission Request (RTR)	
Reserved Bit	1
Data Length Code (DLC)	4
Data Field	0 – 64
CRC	15
ACK	1
End-Of-Frame (EOF)	7

CAN – Devlopment Boards

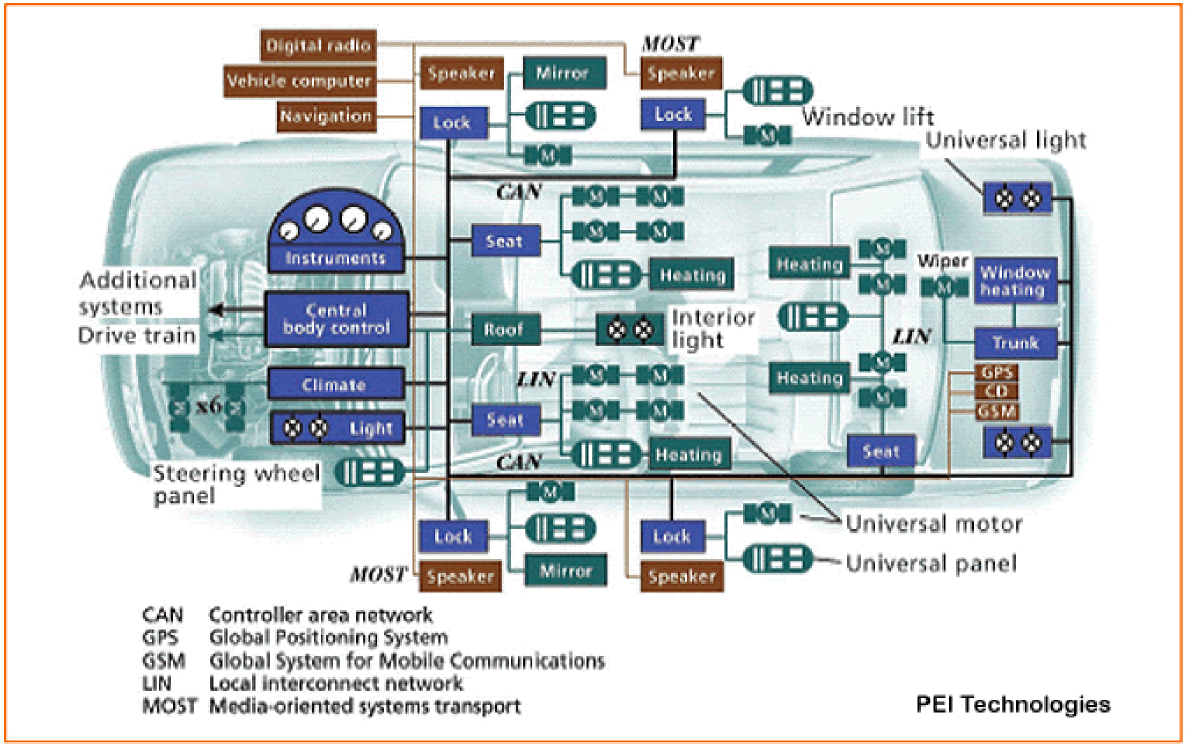
- You can develop CAN applications using simple microprocessor boards interfaced to CAN chipsets.
- You can also use PCI-based CAN HW to develop PC-based applications.





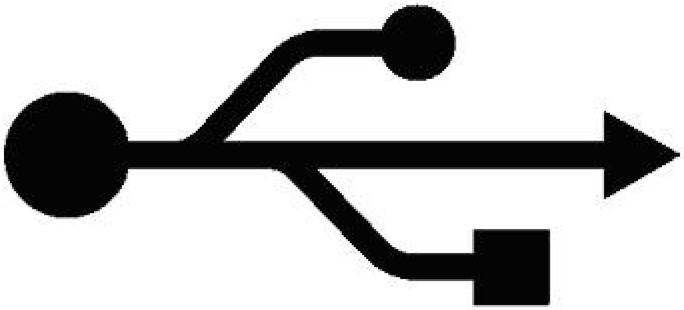
CAN – Applications

- Though CAN is used extensively in automotive applications, it is also used in other areas like
  - Rail and Train networks
  - Aerospace Applications
  - Lifts, Escalators
  - Operating Theatre Equipment
  - Manufacturing Processes, etc..



USB

- Universal Synchronous Bus
  - Set of interface specifications for high speed wired communication between electronics systems peripherals and devices with or without a PC/computer.
  - Major goal was to define an external expansion bus to add peripherals to a PC in an easy and simple manner.
  - Supports Hot-Swapping where devices can be plugged and unplugged without rebooting the computer or turning off the device.
  - Loading of appropriate drivers is done using a PID/VID (Product ID/Vendor ID) combination.

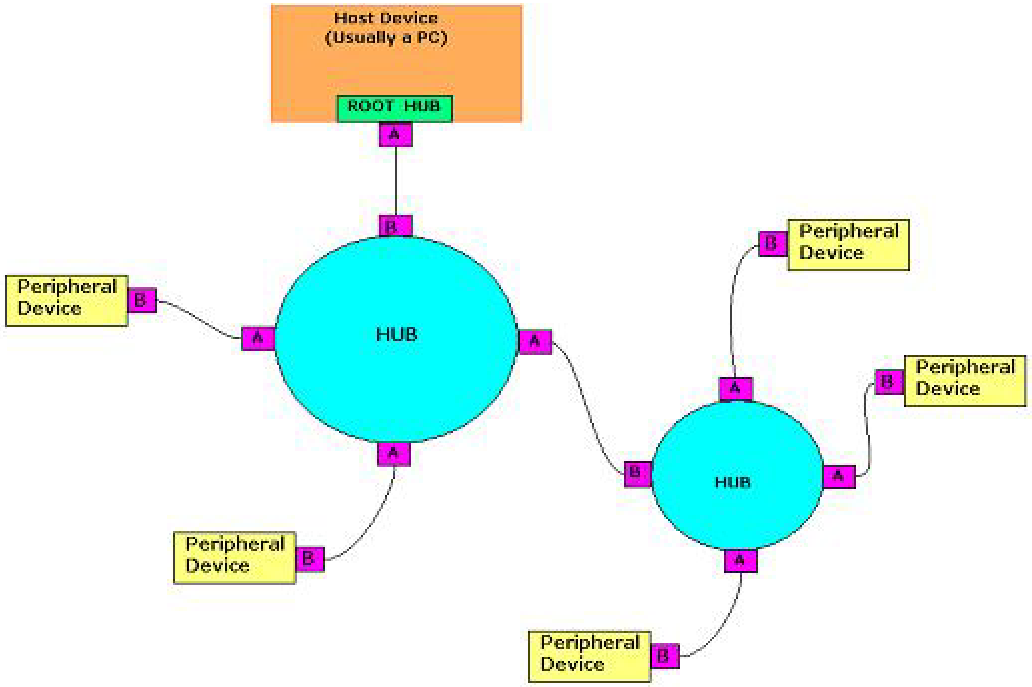


USB - Versions

- USB1.0
  - Launched in January 1996
  - Data rate up to 12Mbps, supporting up to 127 devices
- USB1.1
  - Launched in September 1998
  - Minor modifications for hardware and the specifications
  - Supported Full-Speed mode at 12Mbps and Low-Speed mode at 1.5Mbps
- USB2.0
  - Released in April 2000, also known as Hi-Speed USB
  - Backward compatible with USB1.1 and USB1.0
  - Supports three speeds modes, 480Mbps, 12Mbps and 1.5Mbps
- USB3.0
  - Latest version, also called Super-Speed USB
  - Data rate up to 4 Gbit/s (more than 10x the current Hi-Speed USB speed)
  - Backward compatible with the earlier protocols

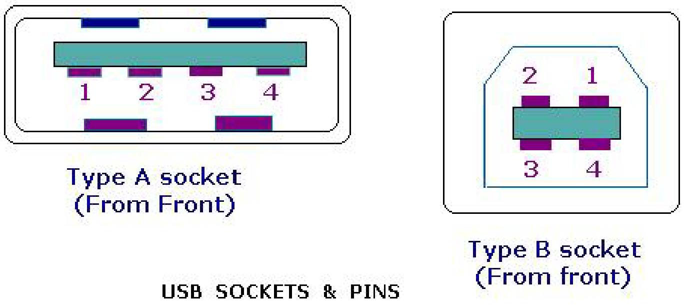
USB – System Overview

- The USB system is made up of a host, multiple number of USB ports, and multiple peripheral devices connected in a tiered star topology.
- Power to each device can be switched off if an overcurrent condition occurs.
- Different speed devices can be supported on the same hub.



USB – Connectors and Power

- The USB standard uses cables with two distinct connectors.
  - Type A – These connectors head upstream toward the computer
  - Type B – These connectors head downstream and connect to individual devices

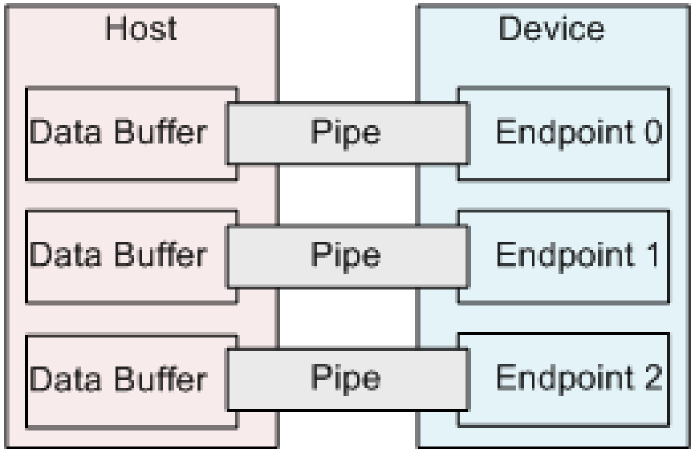


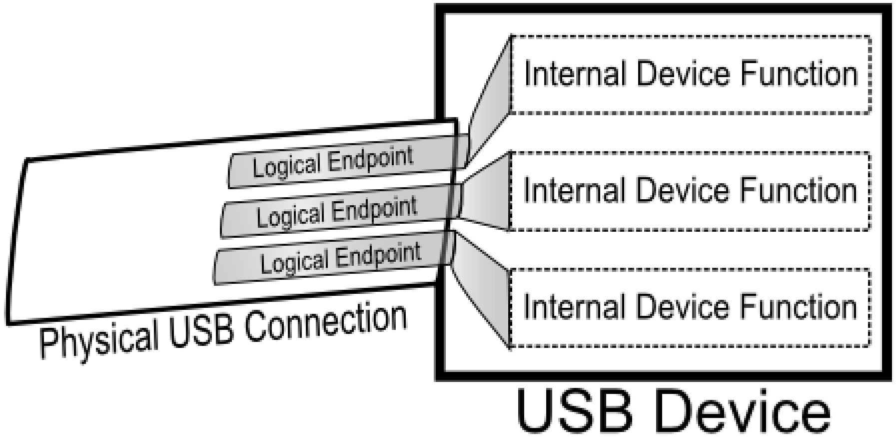
- Inside the cable, there are two wires to supply power and a twisted pair for the data.
- On the power wires, the computer can supply up to 500mA of power at 5Volts.
- If a peripheral device needs more than that, then it needs its own power supply

Pin No.	Signal	Color
1	+ 5V Power	Red
2	- Data	White/Yellow
3	+ Data	Green / Blue
4	Ground	Black / Brown

USB – Communication

- When a USB peripheral device is first attached to the network, a process called enumeration gets started.
- During enumeration, the device information is acquired and it is assigned a unique 7-bit address.
- Communication actually takes place between the host and endpoints located in the peripherals.
- An endpoint is a uniquely addressable portion of the peripheral that is the source or receiver of data.
- Four bits define the devices endpoint address.





USB – Communication

- USB can support four data transfer types (transfer modes)
  - Control
    - Used to exchange configuration, setup and command information between the device and the host.
  - Isochronous
    - Used by time critical, streaming devices such as speakers and video cameras.
    - It has guaranteed access to the USB bus.
  - Bulk
    - Used by devices like printers and scanners, where data is received in big packets.
    - Timely delivery is not critical.
    - Claims unused USB bandwidth when nothing more important is going on.
  - Interrupt
    - Used by peripherals for exchanging small amounts of data that need immediate attention like a mouse or keyboard.

Wireless Connectivity Options

- ZigBee
  - Wireless Technology developed as an open global standard to address the unique needs of low-cost, low-power wireless networks
  - Operates on the IEEE 802.15.4 specifications
  - Protocol created and ratified by members of the ZigBee Alliance
  - Provides a host of monitoring and control applications

- Bluetooth
  - Provides a short-range communication means
  - Operates on the IEE 802.15.1 specifications
  - Robust, low-power and low-cost
  - Provides high data rates

- WiFi
  - Provides High Data Rates
  - Operates on the 802.11 a/b/g/n specifications
  - Used extensively in Building networks
  - Complements Wired Networks